Calculus II: For Physical Sciences and Engineering (AS.110.109)

Lecture: 35, 36

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Textbook Sections: Braun 2.1, 2.2

Today's Topic: Solving 2nd order, linear, homogeneous ODEs with constant coefficients

First, a recap of our setup from last time:

1) We introduced the concept of a linear differential operator: a "function of function" mapping a given function $y(t)$ to a new function $L[y](t)$ built out of $y(t)$ and its derivatives while satisfying the rules

- $L[y_1(t) + y_2(t)] = L[y_1](t) + L[y_2](t)$
- $L[cy(t)] = cL[y](t)$

for all functions y_1, y_2 and all scalars $c \in \mathbb{R}$

We should think of $L[-]$ as a generalization of the derivative operators $\frac{d}{dt}[-]$, $\frac{d^2}{dt^2}[-]$, etc

2) Using this language, by definition a linear ODE can be written as an equation of the form

$$L[y](t) = g(t)$$

where $L[-]$ is a given differential operator, $g(t)$ is a given function, and $y = y(t)$ is the unknown function which we want to solve for. If $g(t) \equiv 0$, $L[y] = 0$ is called a homogeneous equation.

Key Example for today: We will think exclusively about 2nd order linear homogeneous ODEs where $L[-]$ is of the form

$$L[-] = a\frac{d^2}{dt^2}[-] + b\frac{d}{dt}[-] + c$$

where $a, b, c \in \mathbb{R}$, $a \neq 0$. Explicitly, we consider eqtns of the form

$$0 \equiv L[y](t) = a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + c \qquad (*)$$

0 function

As an exercise, check that $L[-]$ is indeed linear.

3) What is the whole point of introducing this operator language?

Writing the ODE (*) using the operator $L[-]$ makes the following points a lot more clear:

(i) If $y_1(t)$ solves (*), then so does $cy_1(t)$ for any $c \in \mathbb{R}$

(ii) If $y_1(t)$ and $y_2(t)$ both solve (*), then so does $y_1(t) + y_2(t)$

Thus, every $y(t) = \alpha y_1(t) + \beta y_2(t)$, $\alpha, \beta \in \mathbb{R}$, solves (*)

(iii) It is possible to find two "independent" solutions $y_1(t), y_2(t)$ to (*) so that every possible solution to (*) is of the form $y(t) = \alpha y_1(t) + \beta y_2(t)$ for some $\alpha, \beta \in \mathbb{R}$.

(iv) Once we have such a "basis set of fundamental solutions," we can uniquely solve every IVP

$$\begin{cases} L[y](t) = y''(t) + p(t)y'(t) + q(t)y(t) = 0 \\ y(0) = y_0 \quad y'(0) = y_0'. \end{cases}$$

by solving for the <u>two</u> unknown coefficients α, β using the <u>two</u> initial conditions.

Key Point: In other words, $L[y]=0$ behaves just like a matrix equation $A\vec{x}=\vec{0}$, where we search for all vectors $x=\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ so that

$$A\vec{x} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0}$$

In this case, the matrix A is a linear operator which maps vectors of some length n to new vectors of length m, and this matrix language thereby gives a way of encoding systems of linear equations. In our ODE case, our unknown function $y(t)$ is like a vector $\vec{x}$, the linear diff. operator $L[-]$ is like the matrix A. In fact, since the space of all functions $y(t)$ is infinite dimensional, $L[y]=0$ is in many senses a sort of infinite dimensional matrix equation! This very special structure/analogy is what leads to the behaviors (i)-(iv) above, and the intuition that $L[-]$ should have "matrix-like" properties ends up being very fruitful!

Let's now solve some equations

We are interested in finding all solutions to the ODE

$$L[y] = ay'' + by' + cy = 0$$

where $a,b,c \in \mathbb{R}$ are fixed and $a \neq 0$. To get a start on finding solutions, first observe that because a,b,c are constant, each of the three terms above ought to be "of similar" type if they are all to cancel out to give the zero function. eg: we can take $y(t)=t^n$ a polynomial, since in this case y'', y', y are all of different orders (and thus can never cancel out in the above equation). We need a function $y(t)$ whose derivatives are all of similar type to $y(t)$ itself.

Idea: Let's try functions like $e^{\lambda t}$, $\cos(\lambda t)$, $\sin(\lambda t)$, etc.

For reasons that will become clear later, it's most efficient to try $e^{\lambda t} = y(t)$ as an ansatz:

$$L[y(t)] = L[e^{\lambda t}] = \left(a\frac{d^2}{dt^2} + b\frac{d}{dt} + c\,Id\right)[e^{\lambda t}]$$

$$= a\lambda^2 e^{\lambda t} + b\lambda e^{\lambda t} + ce^{\lambda t}$$

$$= (a\lambda^2 + b\lambda + c)e^{\lambda t}$$

So: $y(t) = e^{\lambda t}$ solves (*) iff $a\lambda^2 + b\lambda + c = 0$. (**)

Equation (**) is called the characteristic equation of the ODE, and it is a quadratic equation whose roots

$$\lambda_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \qquad \lambda_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

are such that

$$L[e^{\lambda_1 t}] = 0 \quad \text{and} \quad L[e^{\lambda_2 t}] = 0.$$

I.e., the functions $y_{1,2}(t) = e^{\lambda_{1,2} t}$ are both solutions of the ODE!

We should be careful about the possible scenarios that could arise:

Case 1: λ_1, λ_2 are distinct, real roots

Case 2: λ_1, λ_2 are complex conjugates

Case 3: $\lambda_1 = \lambda_2$ is the same, repeated real root.

} On Friday!

Today lets just focus on Case 1. In this case, we obtain two solutions $y_1(t)$ and $y_2(t)$ to the ODE (*). By linearity, we see that

$$y(t) = \alpha e^{\lambda_1 t} + \beta e^{\lambda_2 t}$$

is a solution for any $\alpha, \beta \in \mathbb{R}$. Does this describe all possible solutions?

Yes! As it turns out, the fact that $\lambda_1 \neq \lambda_2$ are distinct implies that $y_1(t) = e^{\lambda_1 t}$ and $y_2(t) = e^{\lambda_2 t}$ are independent solutions (i.e. one is not a scalar multiple of the other). This is exactly what is needed to show that if $y(t)$ is *any* solution of (*), then there must exist α and β s.t. $y(t) = \alpha y_1(t) + \beta y_2(t)$.

(Challenge: Prove this using the uniqueness theorem from last class)

Examples

(1) Find the general soln to $y'' + 5y' + 4y = 0$.

Soln: The associated characteristic equation is $\lambda^2 + 5\lambda + 4 = 0$, which has the distinct real roots $\lambda_1 = 1$, $\lambda_2 = 4$. Thus, the general solution is of the form

$$y(t) = \alpha e^{t} + \beta e^{4t} \qquad \alpha, \beta \in \mathbb{R}.$$

(2) Find the soln of the IVP $\begin{cases} 2y'' + 8y' - 4y = 0 \\ y(0) = 1 \quad y'(0) = 2 \end{cases}$

Soln: Here the characteristic equation is $2\lambda^2 + 8\lambda - 4 = 0$ which has roots $\lambda_1 = -2 - \sqrt{6}$, $\lambda_2 = -2 + \sqrt{6}$ so that the general soln is

$$y(t) = \alpha e^{(-2-\sqrt{6})t} + \beta e^{(-2+\sqrt{6})t}$$

If $y(0) = 1$, then $1 = y(0) = \alpha + \beta$.

If $y'(0) = 2$, then $2 = y'(0) = (-2-\sqrt{6})\alpha + (-2+\sqrt{6})\beta$

and so $\alpha = \frac{1}{2} - \frac{2}{\sqrt{6}}$, $\beta = \frac{1}{2} + \frac{2}{\sqrt{6}}$ give the soln to the IVP.

Lecture 36:

Todays Topic: Cases 2 and 3.

As a quick recap:

We are trying to find all the functions $y = y(t)$ which solve the 2nd order, linear, homogeneous, constant coefficient ODE

$$0 = L[y] = ay'' + by' + cy \qquad (*)$$

Step 1: By looking at the form of the equation, we guessed that a solution $y(t)$ should have the property that its derivatives all resemble one another (as $(*)$ forces them to cancel out). So, we try to find solns of the form $y = y(t) = e^{\lambda t}$.

Step 2: If we feed $y(t) = e^{\lambda t}$ to $L[-]$, we discover the characteristic equation of $(*)$:

$$L[e^{\lambda t}] = (a\lambda^2 + b\lambda + c)\, e^{\lambda t}$$

Thus, $e^{\lambda t}$ solves $L[y] = 0$ iff $a\lambda^2 + b\lambda + c = 0$.

Step 3: We solve the char. eqtn for these values of λ:

$$\lambda_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

Step 4: Now we have cases:

(Lecture 35) **Case 1:** λ_1, λ_2 are real, and have $\lambda_1 \neq \lambda_2$.

In this case, $y_1(t) = e^{\lambda_1 t}$ and $y_2(t) = e^{\lambda_2 t}$ are two "independent" solutions of $L[y] = 0$, and general theory tells us that <u>any other solution</u> of $(*)$ must be of the form

$$y(t) = \alpha y_1(t) + \beta y_2(t)$$

for some choice of constants α, β.

(Lecture 36) Case 2: λ_1, λ_2 are complex conjugates

Case 3: $\lambda_1 = \lambda_2$ is a repeated real root.

Case 2: In this case, the quadratic equation tells us that

$$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

where $b^2 - 4ac < 0$. Let's factor out $\sqrt{-1} = i$:

$$\lambda = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-b \pm \sqrt{(-1)(4ac-b^2)}}{2a} = -\frac{b}{2a} \pm i\frac{\sqrt{4ac-b^2}}{2a}.$$

Thus, $\lambda_1 = \mu - i\xi$ and $\lambda_2 = \mu + i\xi$

for real numbers

$$\mu = -\frac{b}{2a} \quad \text{and} \quad \xi = \frac{\sqrt{4ac-b^2}}{2a}.$$

Therefore: On a (currently formal) level, we have two "independent" solutions to (*):

$$\begin{cases} y_1(t) = e^{\lambda_1 t} = e^{(\mu - i\xi)t} = e^{\mu t} e^{-i\xi t} \\ y_2(t) = e^{\lambda_2 t} = e^{(\mu + i\xi)t} = e^{\mu t} e^{i\xi t} \end{cases}$$

This might be unsatisfying, because we haven't yet discussed whether or not it makes sense to write down the exponential of a complex number. On top of that, it would also be nice to have real valued functions as the solutions of our ODE with real valued coefficients.

Big Idea: Euler's identity: If $x \in \mathbb{R}$, $e^{ix} = \cos x + i \sin x$.

Proof: Using the arithmetic properties of complex numbers, we can (justifiably) compute (using $i=\sqrt{-1}$, $i^2=-1$, $i^3=-i$, $i^4=1$, $i^5=i$, ...) that

$$e^{ix} := \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!} i^n = \underbrace{\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} i^{2n}}_{\text{evens}} + \underbrace{\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} i^{2n+1}}_{\text{odds}}$$

$$= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} (-1)^n + \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} (-1)^n \cdot i$$

$$= \cos x + i \sin x .$$

Using this identity, we thus have that

$$\begin{cases} y_1(t) = e^{\lambda_1 t} = e^{\mu t} e^{-i\xi t} = e^{\mu t}(\cos(\xi t) - i\sin(\xi t)) \\ y_2(t) = e^{\lambda_2 t} = e^{\mu t} e^{i\xi t} = e^{\mu t}(\cos(\xi t) + i\sin(\xi t)) \end{cases}$$

In other words $y(t) = e^{\lambda t} = \underbrace{e^{\mu t}\cos(\xi t)}_{\text{real part of } y(t)} \pm \underbrace{i e^{\mu t}\sin(\xi t)}_{\text{imaginary part of } y(t)}$ solves (*).

This gets us closer to understanding what $e^{\lambda t}$ means when $\lambda \in \mathbb{C}$, but it still doesn't give us $\mathbb{R}$-valued solutions to our $\mathbb{R}$-valued problem. To that end, we again use linearity (i is a scalar after all!):

$$0 = L[e^{\lambda t}] = L[e^{\mu t}\cos(\xi t) \pm i e^{\mu t}\sin(\xi t)]$$

(*) Euler's Identity

linearity

$$= \underbrace{L[e^{\mu t}\cos(\xi t)]}_{\mathbb{R} \text{ part of } L[e^{\lambda t}]} \pm i \underbrace{L[e^{\mu t}\sin(\xi t)]}_{\mathrm{Im} \text{ part of } L[e^{\lambda t}]}$$

Fact: The only way a complex function can be 0 is if both its $\mathbb{R}$ and Im parts are 0! Thus, we must also have that

$$\tilde{y}_1(t) = e^{\mu t}\cos(\xi t) \quad \text{and} \quad \tilde{y}_2(t) = e^{\mu t}\sin(\xi t)$$

are solutions of (*). In fact, these are both "independent" solutions of (*), so as before,

general theory (the Existence/Uniqueness Theorem) tells us that any solution of (*) can be written as

$$y(t) = \alpha \cdot \tilde{y}_1(t) + \beta \tilde{y}_2(t)$$

for some choice of $\alpha, \beta \in \mathbb{R}$. Thus, we can still find $\mathbb{R}$-valued solutions to our ODE problem.

Remark: Solutions of this form model oscillation (this comes from the sine/cosine terms). The oscillations can also exhibit properties like resonance (if $\mu > 0$), or damping (if $\mu < 0$).

Example:

(1) Find the general soln to the ODE

$$4y''(t) + 4y'(t) + 5y(t) = 0$$

Soln: The char eqtn $4\lambda^2 + 4\lambda + 5 = 0$ has roots

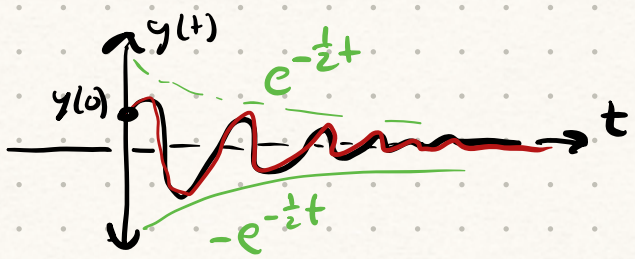
$$\lambda_1 = -\tfrac{1}{2} - i \quad \text{and} \quad \lambda_2 = -\tfrac{1}{2} + i.$$

Thus, we have two indpndt solutions

$$y_1(t) = e^{-\frac{1}{2}t}\cos(t) \quad \text{and} \quad y_2(t) = e^{-\frac{1}{2}t}\sin(t)$$

and the general soln looks like

$$y(t) = \alpha y_1(t) + \beta y_2(t)$$



(2) Find the solution of the IVP

$$\begin{cases} 4y''(t) + 4y'(t) + 5y(t) = 0 \\ y(0) = 1 \quad y'(0) = 1 \end{cases}$$

Soln: From above, the gen. soln looks like

$$y(t) = \alpha e^{-\frac{1}{2}t}\cos t + \beta e^{-\frac{1}{2}t}\sin t$$

for $\alpha, \beta \in \mathbb{R}$.

Applying the initial conditions results in the eqtns

$$1 = y(0) = \alpha \cdot e^0 \cdot \cos(0) + \beta e^0 \cdot \sin(0) = \alpha$$

$$1 = y'(0) = \left(e^{-\frac{1}{2}t}\cos t + \beta e^{-\frac{1}{2}t}\sin t\right)'(0)$$

$$= -\tfrac{1}{2}e^{-\frac{1}{2}t}\left(\cos t + \beta \sin t\right)\Big|_{t=0} + e^{-\frac{1}{2}t}\left(-\sin t + \beta \cos t\right)\Big|_{t=0}$$

$$= -\tfrac{1}{2} + \beta \quad \Rightarrow \quad \beta = 3/2$$

Thus the soln of the IVP is

$$y(t) = e^{-\frac{1}{2}t}\cos t + \frac{3}{2}e^{-\frac{1}{2}t}\sin t$$



Case 3: $\lambda_1 = \lambda_2 = \dfrac{-b}{2a}$ is a repeated real root.

ODDs one, we won't get to this in class...

In any case, we have one soln $y_1(t) = e^{\lambda_1 t} = e^{-\frac{b}{2a}t}$, but need one more independent solution to find the general solution to the ODE.

The ansatz: Try to find a new solution using the old one.

i.e. try to find a function $v(t)$ so that

$$y_2(t) = v(t) \cdot y_1(t)$$

solves (*).

In the case at hand, plugging $y_2(t)$ into $L[\cdot]$ and solving for $v(t)$ gives $v(t) = t$. Thus, in this case the gen. soln is:

$$y(t) = \alpha e^{-\frac{b}{2a}t} + \beta \cdot t e^{-\frac{b}{2a}t}.$$